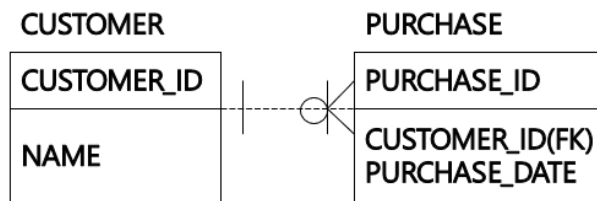


SQLD 서브쿼리 실전편

유형1. 테이블 관계 해석

문제1. 아래 SQL중 실행 결과가 다른 것은?



①

```

SELECT C.CUSTOMER_ID
  FROM CUSTOMER C
 WHERE C.CUSTOMER_ID IN (SELECT P.CUSTOMER_ID
                        FROM PURCHASE P)
 ORDER BY C.CUSTOMER_ID;
  
```

②

```

SELECT C.CUSTOMER_ID
  FROM CUSTOMER C
 WHERE EXISTS (SELECT 1
              FROM PURCHASE P
              WHERE P.CUSTOMER_ID = C.CUSTOMER_ID)
 ORDER BY C.CUSTOMER_ID;
  
```

③

```

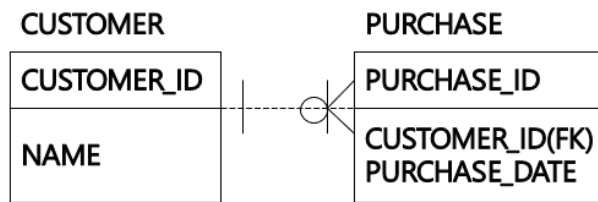
SELECT C.CUSTOMER_ID
  FROM CUSTOMER C JOIN PURCHASE P ON P.CUSTOMER_ID = C.CUSTOMER_ID
 GROUP BY C.CUSTOMER_ID
 ORDER BY C.CUSTOMER_ID;
  
```

④

```

SELECT C.CUSTOMER_ID
  FROM CUSTOMER C JOIN PURCHASE P
    ON P.CUSTOMER_ID = C.CUSTOMER_ID
 ORDER BY C.CUSTOMER_ID;
  
```

문제2. 아래 SQL중 실행 결과가 다른 것은?



①

```

SELECT C.CUSTOMER_ID
  FROM CUSTOMER C LEFT JOIN PURCHASE P ON P.CUSTOMER_ID = C.CUSTOMER_ID
 WHERE P.CUSTOMER_ID IS NULL
 ORDER BY C.CUSTOMER_ID;
  
```

②

```

SELECT C.CUSTOMER_ID
  FROM CUSTOMER C
 WHERE NOT EXISTS (SELECT 1
                    FROM PURCHASE P
                    WHERE P.CUSTOMER_ID = C.CUSTOMER_ID)
 ORDER BY C.CUSTOMER_ID;
  
```

③

```

SELECT DISTINCT C.CUSTOMER_ID
  FROM CUSTOMER C JOIN PURCHASE P ON C.CUSTOMER_ID != P.CUSTOMER_ID;
  
```

④

```

SELECT C.CUSTOMER_ID
  FROM CUSTOMER C
 WHERE C.CUSTOMER_ID NOT IN (SELECT P.CUSTOMER_ID
                             FROM PURCHASE P);
  
```

유형2. 최대/최소 출력

문제1. 아래 SQL중 실행 결과가 다른 것은?

```
CREATE TABLE EMP (
    EMPNO    NUMBER PRIMARY KEY,
    ENAME    VARCHAR2(20),
    JOB      VARCHAR2(20),
    HIREDATE DATE,
    SAL      NUMBER,
    DEPTNO   NUMBER
);
```

①

```
SELECT DEPTNO
FROM EMP
WHERE SAL IN (SELECT MAX(SAL)
              FROM EMP
              GROUP BY DEPTNO)
GROUP BY DEPTNO;
```

②

```
SELECT DEPTNO
FROM EMP
GROUP BY DEPTNO
HAVING MAX(SAL) >= ALL (SELECT MAX(SAL)
                       FROM EMP
                       GROUP BY DEPTNO);
```

③

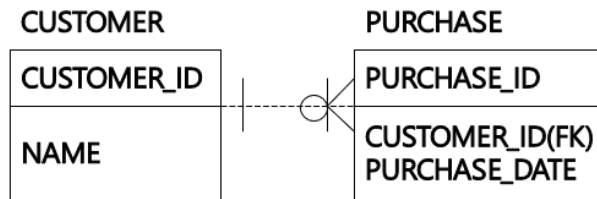
```
SELECT DEPTNO
FROM (SELECT DEPTNO, MAX(SAL) AS MAX1, MAX(MAX(SAL)) OVER() AS MAX2
      FROM EMP
      GROUP BY DEPTNO)
WHERE MAX1 = MAX2;
```

④

```
SELECT DEPTNO
FROM (SELECT DEPTNO, MAX(SAL) AS DEPT_MAX
      FROM EMP
      GROUP BY DEPTNO)
ORDER BY DEPT_MAX DESC
FETCH FIRST 1 ROWS WITH TIES;
```

FINAL. 실전문제

문제1. 아래 SQL중 실행 결과가 다른 것은?



①

```

SELECT C.CUSTOMER_ID,
       C.CUSTOMER_NAME,
       MAX(P.PURCHASE_DATE) AS LAST_PURCHASE_DATE
FROM CUSTOMER C LEFT JOIN PURCHASE P ON P.CUSTOMER_ID = C.CUSTOMER_ID
GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
ORDER BY C.CUSTOMER_ID;
  
```

②

```

SELECT C.CUSTOMER_ID,
       C.CUSTOMER_NAME,
       MAX(P.PURCHASE_DATE) OVER (PARTITION BY C.CUSTOMER_ID) AS LAST_PURCHASE_DATE
FROM CUSTOMER C LEFT JOIN PURCHASE P ON P.CUSTOMER_ID = C.CUSTOMER_ID
ORDER BY C.CUSTOMER_ID;
  
```

③

```

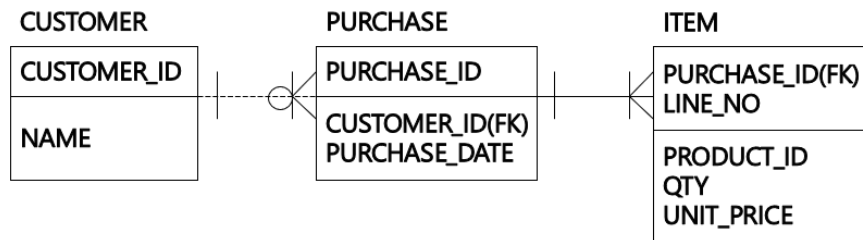
SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, I.LAST_PURCHASE_DATE
FROM CUSTOMER C LEFT JOIN (SELECT CUSTOMER_ID,
                                   MAX(PURCHASE_DATE) AS LAST_PURCHASE_DATE
                           FROM PURCHASE
                           GROUP BY CUSTOMER_ID) I
ON I.CUSTOMER_ID = C.CUSTOMER_ID
ORDER BY C.CUSTOMER_ID;
  
```

④

```

SELECT C.CUSTOMER_ID,
       C.CUSTOMER_NAME,
       (SELECT MAX(P.PURCHASE_DATE)
        FROM PURCHASE P
        WHERE P.CUSTOMER_ID = C.CUSTOMER_ID) AS LAST_PURCHASE_DATE
FROM CUSTOMER C
ORDER BY C.CUSTOMER_ID;
  
```

문제2. 아래 SQL중 실행 결과가 다른 것은?



①

```

SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME,
       COUNT(DISTINCT I.PRODUCT_ID) AS DISTINCT_PRODUCT_CNT
FROM CUSTOMER C LEFT JOIN PURCHASE P ON P.CUSTOMER_ID = C.CUSTOMER_ID
      LEFT JOIN ITEM I ON I.PURCHASE_ID = P.PURCHASE_ID
GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
ORDER BY C.CUSTOMER_ID;
  
```

②

```

SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME,
       COUNT(X.PRODUCT_ID) AS DISTINCT_PRODUCT_CNT
FROM CUSTOMER C LEFT JOIN (SELECT P.CUSTOMER_ID, I.PRODUCT_ID
                           FROM PURCHASE P JOIN ITEM I ON I.PURCHASE_ID = P.PURCHASE_ID
                           GROUP BY P.CUSTOMER_ID, I.PRODUCT_ID) X
ON X.CUSTOMER_ID = C.CUSTOMER_ID
GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
ORDER BY C.CUSTOMER_ID;
  
```

③

```

SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME,
       COUNT(X.PRODUCT_ID) AS DISTINCT_PRODUCT_CNT
FROM CUSTOMER C LEFT JOIN (SELECT P.CUSTOMER_ID, I.PRODUCT_ID
                           FROM PURCHASE P JOIN ITEM I
                           ON I.PURCHASE_ID = P.PURCHASE_ID) X
ON X.CUSTOMER_ID = C.CUSTOMER_ID
GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
ORDER BY C.CUSTOMER_ID;
  
```

④

```
SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME,  
       (SELECT COUNT(*)  
        FROM (SELECT I.PRODUCT_ID  
              FROM PURCHASE P JOIN ITEM I ON I.PURCHASE_ID = P.PURCHASE_ID  
              WHERE P.CUSTOMER_ID = C.CUSTOMER_ID  
              GROUP BY I.PRODUCT_ID)) AS DISTINCT_PRODUCT_CNT  
FROM CUSTOMER C  
ORDER BY C.CUSTOMER_ID;
```

유형1

문제1. ④

문제2. ③

유형2

문제1. ①

실전편

문제1. ②

문제2. ③